

Introducing pharo-agentic-browser

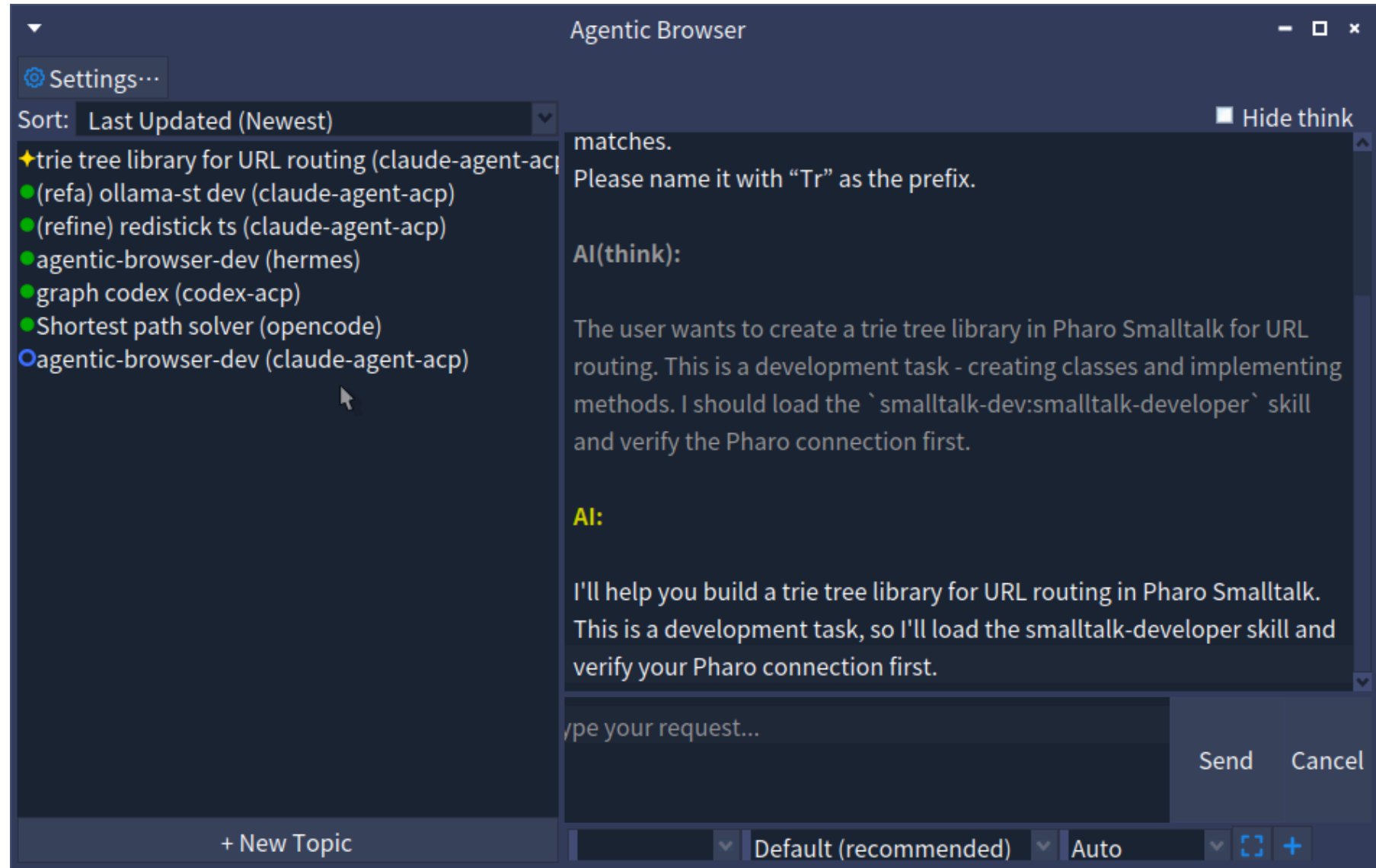
Multi-Agent Session Manager for Pharo Smalltalk

Masashi Umezawa

<https://github.com/mumez/pharo-agentic-browser>

What is pharo-agentic-browser?

An **Integrated AI coding tool** that allows you to use multiple AI coding agents — Claude Code, Codex, OpenCode, and others — from within Pharo.



Each coding session is managed as a **topic**:

1. Create a topic and select an ACP-compatible agent
2. Type a request + mention code with `@ClassName` or attach a screenshot
3. The AI works autonomously (task decomposition, code changes, tests)
4. When the AI needs approval, it pauses and asks in the chat
5. Respond with pulldown selection — the AI resumes
6. Topic status is always visible in the sidebar

Motivation

AI GUI Tools Are Going Mainstream

7

Dedicated GUI tools for AI coding agents are becoming standard:

- **Claude Desktop** — Claude with tools, MCP, and file access
- **Codex Desktop** — OpenAI's autonomous coding environment
- **Cursor / Antigravity / Kiro** — AI-native editors

These tools lower the barrier for interacting with AI agents beyond simple chat.

The Shift: IDE → Multi-Agent Delegation

Era	Paradigm	Examples
Early Days	Chat in editor sidebar	ChatGPT, GitHub Copilot
Little While Ago	AI-first IDE with agent mode	Cursor, Windsurf
Now	Delegate full tasks to agents	Antigravity, Codex Desktop

The AI can now handle **entire feature-level tasks**, not just line completions.

Tools are evolving: the UI is no longer "editor + chat" — it's **session orchestration**.

Pharo developers should be able to leverage the same paradigm:

- **Rich live environment** — classes, methods, and runtime are right there
- **pharo-acp** already provides ACP client support for multiple agents
- **Multiple projects** can run in parallel in one image

A Pharo-native GUI that can control multiple AI agents is the natural next step.

Key Advantages Over Generic Tools

10

Advantage	Details
Direct context passing	<code>@ClassName</code> , <code>@Class>>method</code> , screenshot — no copy-paste
Agent-agnostic	Works with any ACP agent via pharo-acp
Pharo-integrated	Tests, code export, change watching all use the live image
Parallel sessions	Run several agents on different topics simultaneously

Installation

Open a Playground in a Pharo 12+ image and evaluate:

```
Metacello new  
  baseline: 'AgenticBrowser';  
  repository: 'github://mumez/pharo-agentic-browser:main/src';  
  load.
```

Then open the browser:

```
AgenticBrowser open.
```

Any ACP-compatible agent - preset list:

Agent	Install
Claude Code	<code>npm install -g @agentclientprotocol/claude-agent-acp</code>
Codex	<code>npm install -g @agentclientprotocol/codex-acp</code>
Gemini CLI	ACP built-in (<code>gemini --acp</code>)
Copilot CLI	ACP built-in (<code>copilot --acp --stdio</code>)
Cursor CLI	ACP built-in (<code>agent acp</code>)

Also supported: OpenCode, Kilo Code, Kiro CLI and more

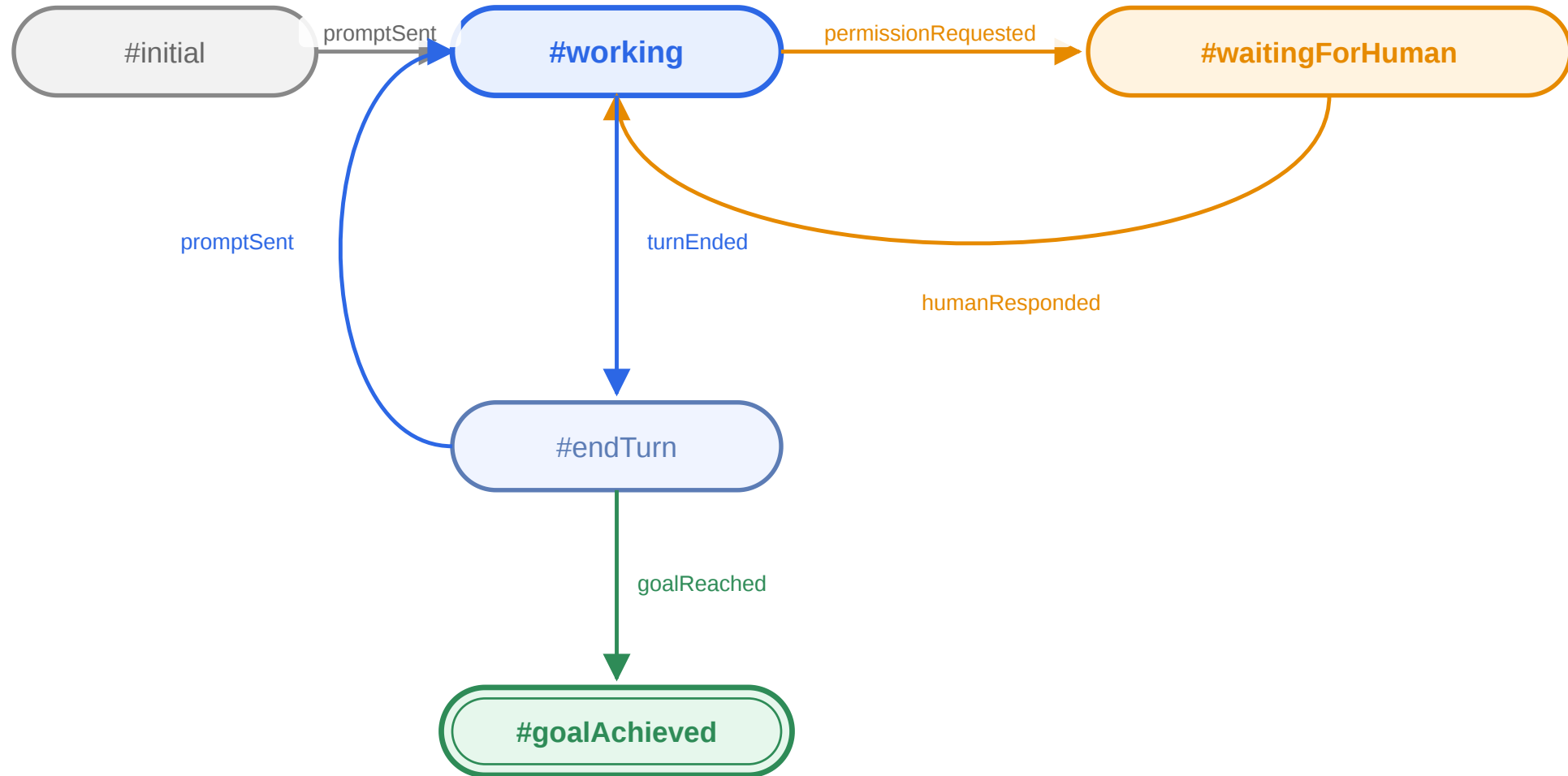
| **Recommended:** Install [smalltalk-dev-plugin](#) in your agent for better Smalltalk

Basic Features

1. Click + **New Topic**
2. Enter a title and select an agent
3. (Optional) Set an existing project directory
4. Click **Create** — topic appears in the left sidebar
5. (Optional) Right-click → **Set Target Packages...** to configure tracked packages

The first message is automatically prefixed with `/st-buddy` to activate the Smalltalk buddy skill.

Each topic follows a clear state machine:



Icon	State	Meaning
	working	AI is actively running
	waitingForHuman	AI needs approval
	endTurn	Turn completed
	goalAchieved	Goal reached

When the AI requests permission:

- Status changes to `?` (waitingForHuman)
- **Send** button becomes **Allow**
- **Cancel** button becomes **Deny**

Respond by clicking the button — the AI resumes seamlessly.

No modal dialogs. Approval is part of the conversation flow.

Reference Pharo classes or methods directly in the chat:

```
@QueryClass @DBAdapter>>connect please refactor this
```

AgenticBrowser resolves each `@mention` to its Tonel source and attaches it as an ACP text resource — **no copy-paste needed**.

Drag and Drop

- Drag a **class** from the System Browser → inserts `@ClassName`
- Drag a **method** from the System Browser → inserts `@ClassName>>methodName`

Click the  button to attach a screenshot:

1. Click the button — cursor becomes a crosshair
2. Drag to select the area
3. A mention like `@sc-20260528-001.png` is inserted in the input
4. Send — the PNG is attached as an image resource to the prompt

Files are saved to `<agenticBrowserRoot>/screenshots/sc-YYYYMMDD-NNN.png`.

Click the + button to attach any file from disk:

1. Click the button — a file selection dialog opens
2. Choose a file — a mention like `[filename]` is inserted in the input
3. Send — the file's contents are attached as a text resource

Oversized files are truncated to `maxAttachmentSize` (default 5MB) before being embedded in the prompt.

Removing the `[filename]` mention text before sending cancels the attachment.

Right-click a topic → **Set Goal...** to enter a completion condition:

```
all tests pass
```

AgenticBrowser sends a goal prompt to the AI:

Goal has been set: all tests pass. When the goal is achieved, summarize and report in result-`<topic-id>`.md. Keep retrying until the goal is achieved.

When `result-<topic-id>.md` is created, the topic transitions to ✓ (`#goalAchieved`).

Two hooks fire when a goal is reached:

Announcement

```
topic announcer
  when: AbTopicGoalAchieved
  do: [:ann | Transcript crShow: ann topic title , ' achieved: ' , ann goal result].
```

Callback Block

```
topic whenGoalAchieved: [:goal | Transcript crShow: goal result].
```

Integrate goal events into your own automation workflows.

Topics are automatically saved to `ab-topics.fuel` using Pharo's **Fuel** serializer — they survive image restarts.

Manual save/restore:

```
AbTopicManager save.  
AbTopicManager load.
```

Per-topic state (settings, status, conversation) is fully persisted.

AgenticBrowser watches the image for edits to a topic's tracked packages:

- Right-click a topic → **Set Target Packages...** to configure prefixes (e.g. `# ('AgenticBrowser-*')`)
- When a matching class/method is saved, a system message appears in chat and the package is exported after confirmation
- Edits to **untracked** packages are collected — promote them via **Apply Updated External Packages**

Keeps the AI's view of the Tonel source in sync with whatever you change live in the image.

Customization

Every new topic's working directory is seeded from `<agenticBrowserRoot>/topic-template`:

- Ships with a default `CLAUDE.md` / `AGENTS.md` tuned for the `smalltalk-dev-plugin`
- Drop in `.claude`, `.opencode`, or other agent config directories — skills, commands, rules shared across all topics
- Replace the defaults with your own customized versions any time

Saves you from reconfiguring the coding agent for every new topic. Not applied when a topic points at an existing project directory.

Place `mcp.json` in your AgenticBrowser root directory:

```
{
  "mcpServers": {
    "my-server": {
      "command": "uvx",
      "args": ["my-server-package"],
      "env": {"API_KEY": "value"}
    }
  }
}
```

- Built-in `smalltalk-interop` and `smalltalk-validator` servers are **auto-merged** by default
- Set `useDefaultMcpServers: false` to use only your own `mcp.json`

From a Playground

```
AbSettings default codingAgents: (AbSettings default codingAgents copyWith:
  {'name' -> 'my-agent'.
  'command' -> #('my-agent' '--acp')} asDictionary).
AbSettings save.
```

Or edit **ab-settings.json** directly

The agent appears as a preset in the **New Topic** dialog on next open.

Key	Default	Description
<code>useDefaultMcpServers</code>	<code>true</code>	Merge built-in Smalltalk MCP servers
<code>aiPermissionWaitTimeoutSeconds</code>	<code>1800</code>	Timeout for human approval
<code>aiPermissionTimeoutOption</code>	<code>#reject_once</code>	Auto-response: <code>allow_once</code> , <code>allow_always</code> , <code>reject_once</code>
<code>exportApprovalWaitTimeoutSeconds</code>	<code>30</code>	Timeout for package export approval

Settings can also be configured **per-topic** via right-click → **Edit Settings...**

Two more ways to work with AgenticBrowser, beyond the Spec UI:

- **Web UI** — web browser interface over WebSocket; mobile friendly, live topic updates
→ [Web UI slides](#)
- **Scripting API** — simple DSL to build and run multi-agent orchestrations from code
(sequential / parallel steps, save & load)
→ [Scripting API slides](#)

Summary

pharo-agentic-browser brings the multi-agent delegation paradigm to Pharo:

- **Native GUI** — manage multiple AI sessions without leaving your image
- **Agent-agnostic** — any ACP-compatible agent works
- **Rich context** — code mentions, drag-and-drop, screen captures
- **Human-in-the-loop** — conversational approval, no interruption dialogs
- **Goal-driven** — set completion conditions, hook into results
- **Extensible** — custom MCP servers and agents with minimal configuration

Feedback and contributions are welcome!

<https://github.com/mumez/pharo-agentic-browser>